Chapter 2: Linear Regression

Predicting Elastic Stress

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Outline

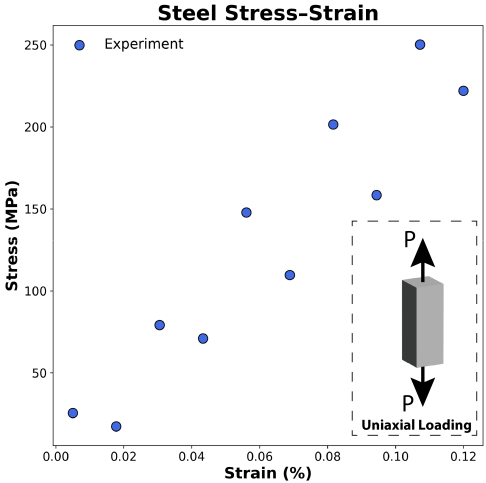
- Problem setup: stress–strain of steel
- Linear regression model
- Loss functions (MSE, MAE)
- Estimating parameters: closed-form and gradient descent
- Generalization and extrapolation
- Overfitting

Stress–Strain of Steel

- Steel specimen under uniaxial tensile load P.
- Engineering strain: $\varepsilon = \Delta L / L_0$.
- Engineering stress: $\sigma = P / A_0$.
- $N = 10$ synthetic measurements (same workflow applies to real experiments).

Goals:

1. Predict stress at unmeasured strains.
2. Estimate the elastic modulus E.



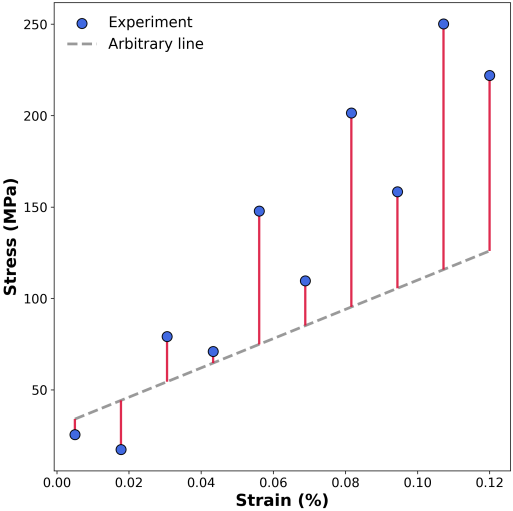
Stress–strain data points.

The Best-Fit Question

Under Hooke's law: $\sigma = E\varepsilon$.

- In practice, measurements include noise.
- A single line cannot pass through every point exactly.
- Each vertical deviation is a **residual**: $r_i = \sigma_i - \hat{\sigma}(\varepsilon_i)$.

Key question: What line provides the best fit?



Residuals between data and a line.

The Linear Model

The predicted stress at strain ε:

$$\hat{\sigma}(\varepsilon) = w_1 \, \varepsilon + w_0$$

Slope w_1 Estimate of the elastic modulus E (units: MPa).

Intercept w_0 Bias term — ideally ≈ 0 for Hooke's law, but left free to capture measurement offsets.

In ML terminology: w_0 and w_1 are **model parameters** learned from data.

Loss Functions: MSE vs. MAE

A **loss function** maps model parameters to a single scalar measuring overall misfit.

Mean Squared Error (MSE): $$J_{\mathrm{MSE}} = \frac{1}{N}\sum_{i=1}^{N}\big(\sigma_i - (w_1\varepsilon_i + w_0)\big)^2$$

Mean Absolute Error (MAE): $$J_{\mathrm{MAE}} = \frac{1}{N}\sum_{i=1}^{N}\big|\sigma_i - (w_1\varepsilon_i + w_0)\big|$$

- MSE penalizes large residuals more strongly (squaring) — sensitive to outliers but smooth.
- MAE is more robust to outliers but not differentiable at zero.
- MSE is the most common choice and is used throughout this course.

Closed-Form Solution

Set $\partial J/\partial w_0 = 0$ and $\partial J/\partial w_1 = 0$, then solve the **normal equations**:

$$w_1 = \frac{\sum_{i=1}^{N}(\varepsilon_i - \bar{\varepsilon})(\sigma_i - \bar{\sigma})}{\sum_{i=1}^{N}(\varepsilon_i - \bar{\varepsilon})^2}, \qquad w_0 = \bar{\sigma} - w_1\,\bar{\varepsilon}$$

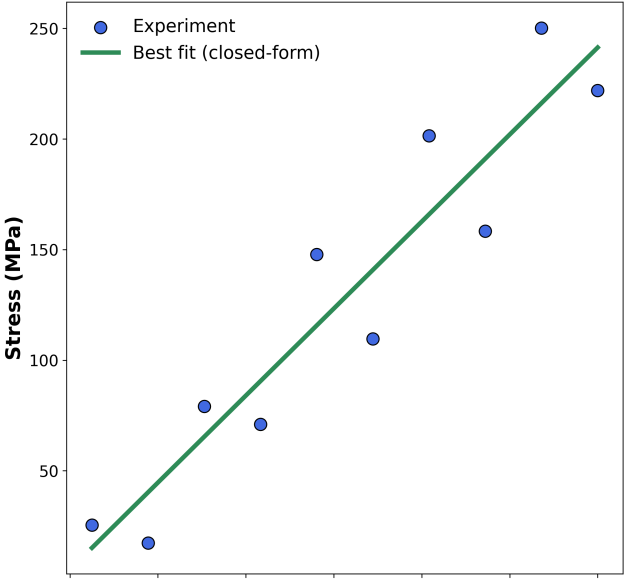
Results for the synthetic data:

$$w_1 \approx 1.97 \times 10^5 \text{ MPa}, \qquad w_0 \approx 5.30 \text{ MPa}$$

The estimated E is $\sim$94% of the manufacturer-specified value (210 GPa).

Limitation: Works only when the model is linear in parameters and the loss is quadratic.

Best-Fit Result



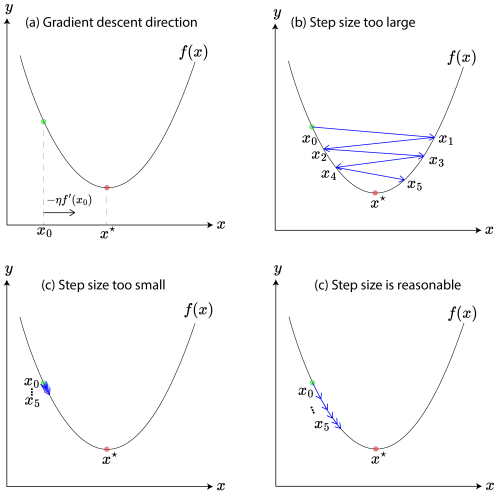
Gradient Descent — Concept

An iterative approach that works for *any* differentiable loss:

$$x_{k+1} = x_k - \eta\, f'(x_k)$$

- $\eta > 0$: **learning rate** (a hyperparameter).
- Move in the direction of the *negative gradient* (downhill).

Hyperparameter vs. parameter:
w_0, w_1 are *learned*; η is *chosen* by the user.



(a) One step. (b) η too small. (c) η too large. (d) Good

Gradient Descent for Linear Regression

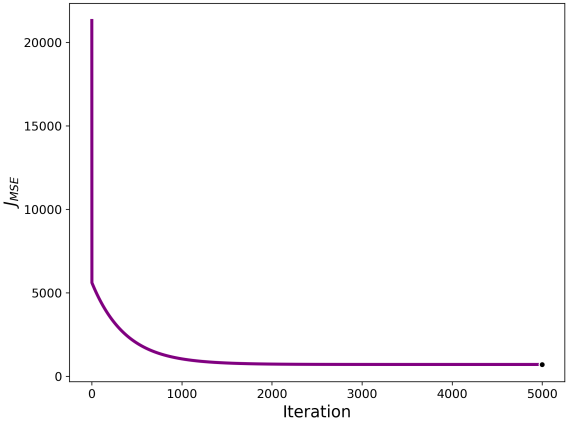
Apply GD to J_{MSE}: $\quad w_0 \leftarrow w_0 - \eta \dfrac{\partial J}{\partial w_0}, \quad w_1 \leftarrow w_1 - \eta \dfrac{\partial J}{\partial w_1}$

Gradients:

$$\frac{\partial J}{\partial w_0} = -\frac{2}{N}\sum_{i=1}^{N}\big(\sigma_i - (w_1\varepsilon_i + w_0)\big), \quad \frac{\partial J}{\partial w_1} = -\frac{2}{N}\sum_{i=1}^{N}\varepsilon_i\big(\sigma_i - (w_1\varepsilon_i + w_0)\big)$$

Starting from $w_0 = 0$, $w_1 = 50$, with $\eta = 0.5$, after 5,000 iterations: $w_1 \approx 1964.9$, $w_0 \approx 5.45$.

Loss Curve



MSE loss J_{MSE} vs. gradient descent iteration.

- Loss decreases monotonically and converges.
- **SGD:** Use a random mini-batch per step — cheaper updates, scales to large datasets.

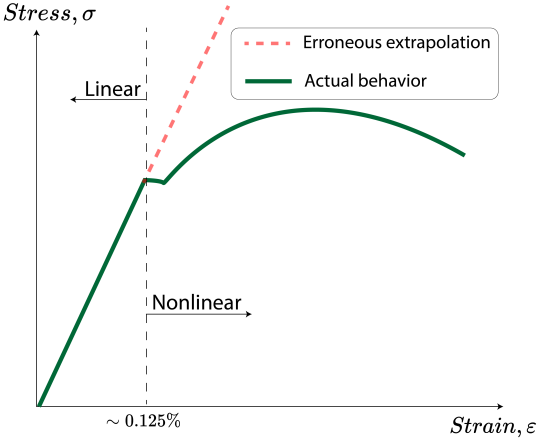
What is Generalization?

Definition

Generalization = performance on *new*, *unseen* samples from the same distribution (in-distribution).

Interpolation	Predicting within the observed data range — higher confidence.
Extrapolation	Predicting beyond the observed range — often unreliable.
Out-of-distribution	Test conditions differ meaningfully from training (e.g. yielding regime).

Extrapolation Can Fail



Erroneous extrapolation beyond the elastic limit.

Validating Generalization

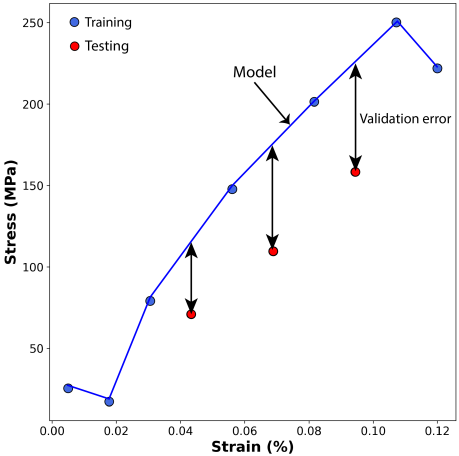
Hold-out validation:

- Split data (e.g. 80/20): fit on 80%, evaluate on 20%.
- Estimates *in-distribution* generalization.

Key principle: The test set must *never* be used during training.

Overfitting: Too Much Complexity

- A model can **memorize** training data rather than learning the trend.
- Example: a piecewise-linear model connecting all N points achieves *zero* training error but generalizes poorly.
- **Occam's razor:** among models with similar performance, prefer the simpler one.



Piecewise-linear model overfits; linear model generalizes better.

Key Takeaways

1. A **loss function** quantifies prediction error (MSE is the default for regression).
2. **Gradient descent** iteratively minimizes the loss — more general than closed-form solutions.
3. The **learning rate** is a critical hyperparameter: too large $\rightarrow$ divergence; too small $\rightarrow$ slow convergence.
4. **Generalization** is the ultimate goal — validate on held-out data.
5. **Overfitting** = fitting noise; prefer simpler models when performance is comparable.